

Structure-Based Prediction of NES Motifs using GNNs

Tzvi Marmor, Amit Yosef, Shaked Hartal, Hadar Dayan
Hebrew University of Jerusalem

Hackathon Presentation – 3D Data Processing in Structural
Biology

Project Goal

Objective: Predict NES (Nuclear Export Signals) in proteins using 3D structural data and Graph Neural Networks.

- ▶ Utilize AlphaFold-predicted PDB files.
- ▶ Construct graphs where nodes = residues, edges = spatial proximity.
- ▶ Use GNNs to classify NES presence.

Data and Preprocessing

- ▶ 675 protein structures (AlphaFold PDBs).
- ▶ Labels: NES-positive / NES-negative.
- ▶ Each residue: C-alpha atom used as graph node.
- ▶ Edges between residues j 8Å apart.

Graph Construction Pipeline

1. Filter to chain B (NES motif) + surrounding residues ($\pm 15\text{\AA}$).
2. Encode residues: One-hot + NES-chain binary.
3. Build PyTorch Geometric graph (x , edge_index, edge_attr, pos).
4. Optional: export filtered PDB and visualize.

Visualizations

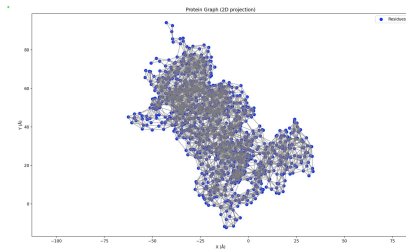


Figure 1: Full dense protein graph.

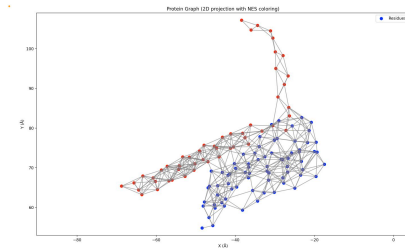


Figure 2: NES chain (red) + nearby residues (blue).

Model Selection: EGNN vs. GVP-GNN

Feature	GVP-GNN	EGNN
Handles vectors?	Yes	No
Equivariance	SE(3)	Rotation/Translation/Reflection
Complexity	High	Moderate
Data Efficiency	Low	High
Best for CA atoms?	No	Yes

Why EGNN?

- ▶ Our graphs only use distances (not full vectors).
- ▶ about 600 examples – EGNN less prone to overfitting.
- ▶ Simpler, faster for experimentation during hackathon.

Model Architecture (EGNN)

- ▶ Node features: amino acid type + NES-chain binary.
- ▶ Edge features: Euclidean distance.
- ▶ Node/pos updated jointly in equivariant fashion.
- ▶ Output: classification (NES present or not).

Baseline Model: GCN (Graph Convolutional Network)

What is GCN?

- ▶ Neural network designed for graph-structured data (like proteins).
- ▶ Acts like CNNs but for graphs — aggregates neighbor features.

How It Works

1. Each node (residue) has a one-hot vector for amino acid type.
2. Edges connect $C\alpha$ atoms within a certain distance.
3. Feature updates via weighted aggregation from neighbors.
4. Global pooling used for whole-protein classification.

Baseline Model: GCN (Graph Convolutional Network)

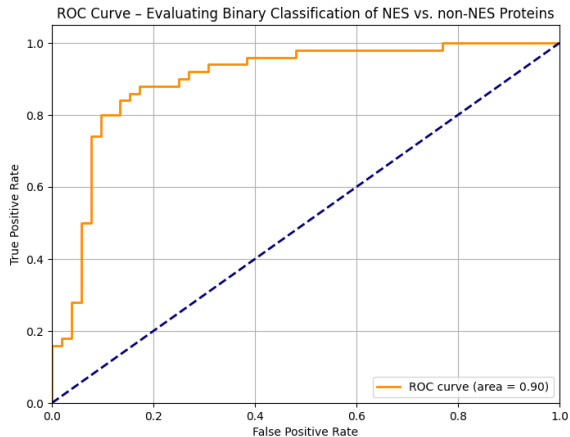
Pros

- ▶ Simple and efficient.
- ▶ Validates preprocessing pipeline.
- ▶ Good for early experimentation.

Cons

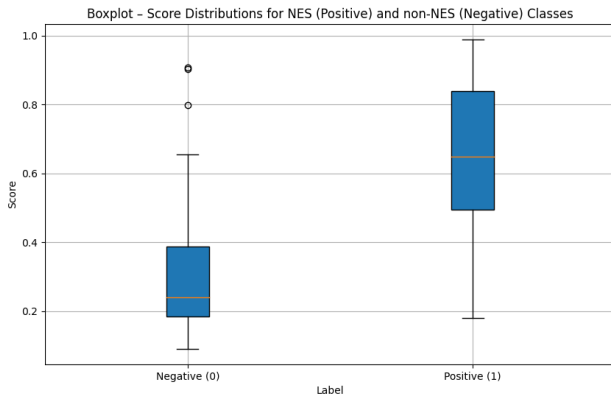
- ▶ Ignores 3D geometry (no distance or positional data).
- ▶ Less expressive than EGNN or GVP-GNN.

Model Performance: ROC Curve



ROC curve visualizing the trade-off between true positive rate and false positive rate. $AUC = 0.90$ indicates excellent discrimination.

Model Performance: Score Distributions



Boxplot showing predicted probability distributions for NES-positive and NES-negative samples.

Future Work

- ▶ Integrate ESM embeddings + pLDDT.
- ▶ Attention visualization on NES pockets.
- ▶ Try SE(3)-Transformer on same graph structure.

Thank You!

Questions? Feedback?

Team: Tzvi Marmor, Amit Yosef, Shaked Hartal, Hadar Dayan